AUDIOSHIELD FORENSICS

Audio Forensic Analysis Report

Report ID	ASF-20260918-EA4F
Generated	2026-09-18 14:15:18 India Standard Time
File analyzed	LA_E_5849185.flac

This report presents separate evidence streams and conservative forensic indicators. It is not a substitute for expert examination of the original evidence.

1. Executive Summary

Overall risk assessment: **low**

Overall assessment: **likely_authentic** | Confidence: **0.760250561777668**

Key findings

- AI consensus: bonafide (risk: low).
- Metadata risk: low; 2 finding(s).
- Signal risk: low; 0 anomaly/anomalies.

Official Eval AI comparison: CNN 69.83%, SVM 89.40%, AASIST 86.45%.

AASIST is the primary forensic model, achieving 99.79% precision and 7.20% EER on the official eval set.

Recommendation: Likely authentic based on the available indicators, subject to further verification.

2. File Information

Property	Value
File name	46fd7184-df87-4a09-a867-1ebfd181f105.flac
File size (bytes)	75249
Format	flac
Duration (seconds)	4.392750
Sample rate (Hz)	16000
Channels	1
Codec	flac
Bitrate (bits/sec)	137042
Creation time	2026:09:17 18:28:25+05:30
Modification time	2026:09:17 18:28:25+05:30

3. AI Detection Results

Official Eval-set comparison

Rank	Model	Accuracy	Precision	EER
1st	AASIST	86.45%	99.79%	0.0720
2nd	SVM	89.40%	95.44%	0.1698
3rd	CNN	69.83%	99.99%	0.0716

Ranking prioritizes forensic suitability: **precision first** (50%), then **EER** (35%), then accuracy (15%). Precision is the primary criterion because a false positive — flagging a genuine speaker as synthetic — carries the highest forensic cost. CNN ranks 3rd despite the highest precision (99.99%) because its recall of only 66.4% means it misses one-third of all spoof samples.

Best model: **AASIST** (best forensic suitability: highest precision 99.79%, low EER 7.20%, strong accuracy 86.45%). AASIST is recommended as the primary model due to its highest precision among well-performing models and the lowest false positive rate (1.54%).

4. Metadata Forensics

Extracted metadata

Field	Value
File Name	46fd7184-df87-4a09-a867-1ebfd181f105.flac
File Size	75249
Format	flac
Duration	4.392750
Codec	flac
Sample Rate	16000
Channels	1
Bitrate	137042
Encoder	Not available
Creation Time	2026:09:17 18:28:25+05:30
Modification Time	2026:09:17 18:28:25+05:30
Device Info	Not available
Software	Not available
Comments	Not available

Risk score: **low**

Findings

Severity	Field	Description
low	device	Device information is missing; provenance cannot be verified from metadata alone.
low	encoder	Encoder or software information is missing.

Metadata indicators are not conclusive on their own and require further investigation.

5. Signal Forensics

Spectral features

Feature	Mean	Std	Min	Max
Spectral Centroid	1220.6410496138178	783.7913829685122	0.0	4683.424306653209
Spectral Bandwidth	1568.0357423438826	700.126634283738	0.0	2919.755865543769
Spectral Rolloff	2769.854697452229	1759.809549945914	0.0	6875.0
Spectral Flatness	0.11836665453760778	0.30787900633507575	0.000147970029502175 75	1.0000007152557373

Feature	Mean	Std	Min	Max
Spectral Contrast	19.43972139053957	4.392050034654948	3.2286318190080943	28.83425701488692

Temporal features

Feature	Mean	Std	Min	Max
Zero Crossing Rate	0.047503234474522295	0.055599724612923104	0.0	0.3115234375
Rms Energy	0.071280622893016021	0.076770723394584521	0.0	0.248046010732650761
Pitch	84.69792099668587	10.157695116219449	68.49972922973525	110.63722351746388

Risk score: **low**

Anomalies

No signal-level anomalies were reported.

6. Evidence Fusion

Evidence source	Risk / result
AI	bonafide (risk: low)
Metadata	low
Signal	low
Overall	low

Assessment: **likely_authentic** | Confidence: **0.760250561777668**

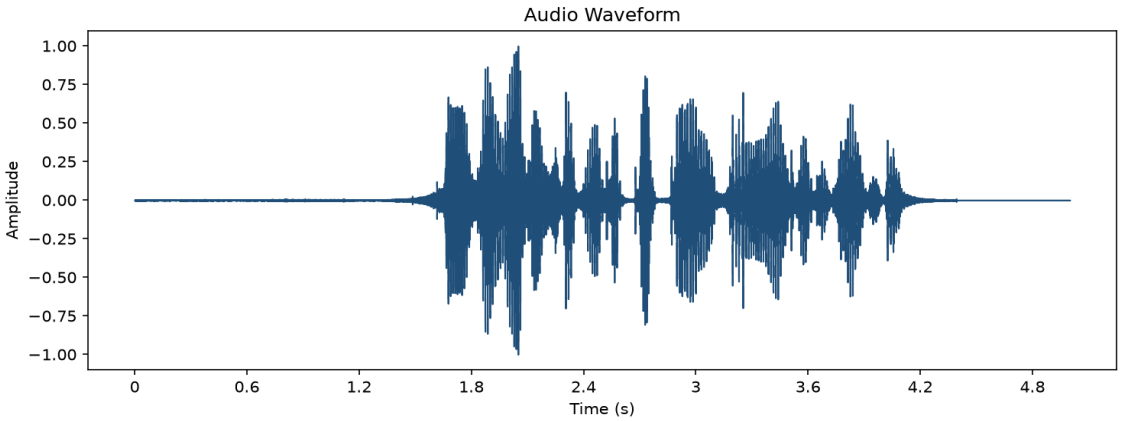
Reasoning: Source risks are AI=low, metadata=low, signal=low; applied documented rule-based logic.

Recommendation: Likely authentic based on the available indicators, subject to further verification.

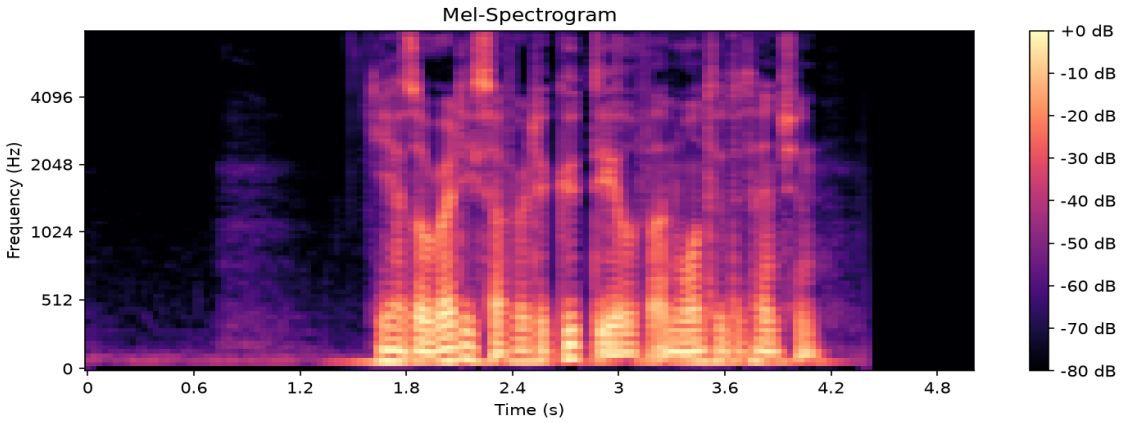
AI model breakdown

Model	Label	Confidence
CNN	bonafide	0.9999999999990312
SVM	spoof	0.999999264908558
AASIST	bonafide	0.28075242042541504

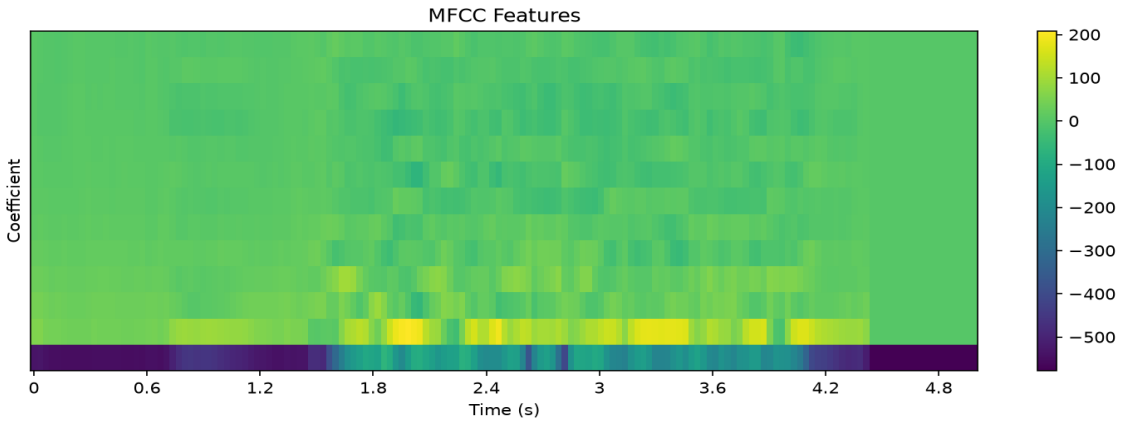
7. Visual Analysis



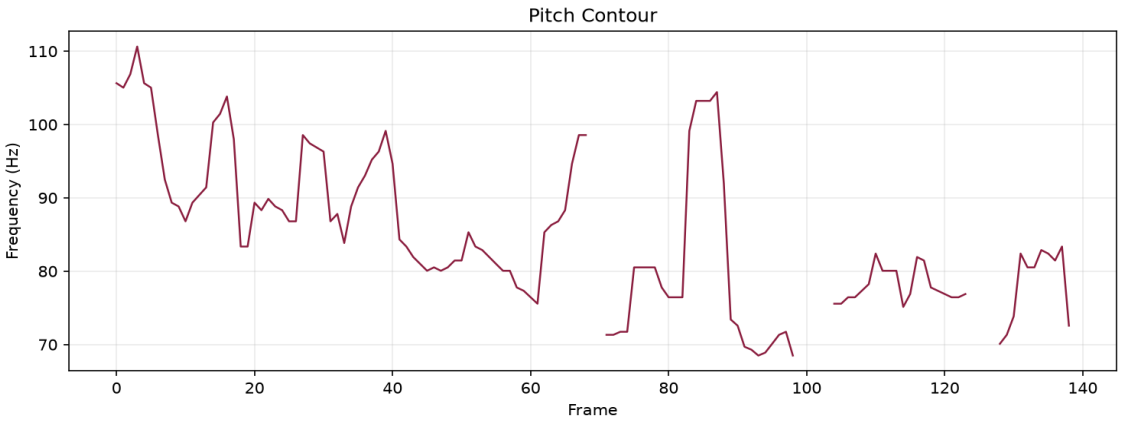
Waveform: Time-domain waveform.



Spectrogram: Frequency content over time.



MFCC: Mel-frequency cepstral coefficients.



Pitch contour: Estimated fundamental frequency.

8. Conclusion

The combined evidence indicates an overall assessment of **likely_authentic** with a reported risk level of **low**. These are indicators from the available reports and should be interpreted with the original audio and chain of custody.

Limitations: AI results may be unavailable; metadata can be edited or lost; signal features are influenced by recording conditions and preprocessing; and this report does not establish authenticity by itself.

9. Appendix

Raw AI evaluation results


```
{
  "CNN": {
    "model": "CNN",
    "checkpoint": "C:\\Users\\allen\\AudioShield-Forensics\\models\\cnn\\cnn_final_best.pth",
    "dataset": "ASVspoof 2019 LA",
    "split": "eval",
    "total_samples": 71237,
    "bonafide_samples": 7355,
    "spoof_samples": 63882,
    "best_epoch": 4,
    "best_validation_loss": 0.003000220499216719,
    "threshold": {
      "value": 0.40700000000000003,
      "source": "frozen_dev_threshold",
      "selection_method": "balanced_accuracy_with_spoof_recall_constraint",
      "selection_source": "dev_only",
      "path": "C:\\Users\\allen\\AudioShield-Forensics\\results\\cnn_final_threshold.json"
    },
    "normalization": {
      "method": "min-max",
      "train_min": -80.0,
      "train_max": 0.0,
      "source": "C:\\Users\\allen\\AudioShield-Forensics\\results\\cnn_final_norm.json"
    },
    "threshold_independent_metrics": {
      "roc_auc": 0.9800478133427984,
      "eer": 0.0716185237520802,
      "eer_percent": 7.161852375208019
    },
    "metrics": {
      "accuracy": 0.6982888105900024,
      "precision": 0.9998820726903936,
      "recall": 0.6636298174759713,
      "f1_score": 0.7977719441856964,
      "specificity": 0.9993201903467029,
      "fpr": 0.0006798096532970768,
      "tpr": 0.6636298174759713,
      "balanced_accuracy": 0.8314750039113371,
      "threshold": 0.40700000000000003,
      "selection_method": "balanced_accuracy_with_spoof_recall_constraint"
    },
    "confusion_matrix": {
      "labels": [
        "bonafide",
        "spoof"
      ],
      "matrix": [
        [
          7350,
          5
        ],
        [
          21488,
          42394
        ]
      ]
    }
  },
  "SVM": {
    "model": "SVM",
    "split": "official_eval",
    "total_samples": 71237,
```

```
    "bonafide_samples": 7355,
    "spoof_samples": 63882,
    "metrics": {
      "accuracy": 0.8940017125931748,
      "precision": 0.9543628708319218,
      "recall": 0.926082464544003,
      "f1_score": 0.9400100102485879,
      "roc_auc": 0.874909277304299,
      "eer": 0.16983042600362058
    }
  },
  "AASIST": {
    "total_samples": 71237,
    "bonafide_samples": 7355,
    "spoof_samples": 63882,
    "tn": 7242,
    "fp": 113,
    "fn": 9540,
    "tp": 54342,
    "accuracy": 0.8644945744486713,
    "precision": 0.9979248921127537,
    "recall": 0.850662158354466,
    "f1": 0.9184278796995022,
    "specificity": 0.9846363018354861,
    "fpr": 0.015363698164513936,
    "tpr": 0.850662158354466,
    "roc_auc": 0.9769212178274564,
    "eer": 0.07204162071337723,
    "threshold": 0.8189939028042915
  }
}
```


Raw metadata report


```
{
  "metadata": {
    "file_name": "46fd7184-df87-4a09-a867-1ebfd181f105.flac",
    "file_size": "75249",
    "format": "flac",
    "duration": "4.392750",
    "codec": "flac",
    "sample_rate": "16000",
    "channels": 1,
    "bitrate": "137042",
    "encoder": null,
    "creation_time": "2026:09:17 18:28:25+05:30",
    "modification_time": "2026:09:17 18:28:25+05:30",
    "device_info": null,
    "software": null,
    "comments": null,
    "raw_ffprobe": {
      "streams": [
        {
          "index": 0,
          "codec_name": "flac",
          "codec_long_name": "FLAC (Free Lossless Audio Codec)",
          "codec_type": "audio",
          "codec_tag_string": "[0][0][0][0]",
          "codec_tag": "0x0000",
          "mime_codec_string": "flac",
          "sample_fmt": "s16",
          "sample_rate": "16000",
          "channels": 1,
          "channel_layout": "mono",
          "bits_per_sample": 0,
          "initial_padding": 0,
          "r_frame_rate": "0/0",
          "avg_frame_rate": "0/0",
          "time_base": "1/16000",
          "start_pts": 0,
          "start_time": "0.000000",
          "duration_ts": 70284,
          "duration": "4.392750",
          "bits_per_raw_sample": "16",
          "extradata_size": 34,
          "disposition": {
            "default": 0,
            "dub": 0,
            "original": 0,
            "comment": 0,
            "lyrics": 0,
            "karaoke": 0,
```

```
            "forced": 0,
            "hearing_impaired": 0,
            "visual_impaired": 0,
            "clean_effects": 0,
            "attached_pic": 0,
            "timed_thumbnails": 0,
            "non_diegetic": 0,
            "captions": 0,
            "descriptions": 0,
            "metadata": 0,
            "dependent": 0,
            "still_image": 0,
            "multilayer": 0
          }
        }
      ],
      "format": {
        "filename": "C:\\Users\\allen\\AudioShield-Forensics\\uploads\\46fd7184-df87-4a09-a867-1ebfd181f105.flac",
        "nb_streams": 1,
        "nb_programs": 0,
        "nb_stream_groups": 0,
        "format_name": "flac",
        "format_long_name": "raw FLAC",
        "start_time": "0.000000",
        "duration": "4.392750",
        "size": "75249",
        "bit_rate": "137042",
        "probe_score": 100
      }
    },
    "raw_exiftool": {
      "SourceFile": "C:/Users/allen/AudioShield-Forensics/uploads/46fd7184-df87-4a09-a867-1ebfd181f105.flac",
      "ExifToolVersion": 13.59,
      "FileName": "46fd7184-df87-4a09-a867-1ebfd181f105.flac",
      "Directory": "C:/Users/allen/AudioShield-Forensics/uploads",
      "FileSize": "75 kB",
      "FileModifyDate": "2026:09:17 18:28:25+05:30",
      "FileAccessDate": "2026:09:17 18:28:35+05:30",
      "FileCreateDate": "2026:09:17 18:28:25+05:30",
      "FilePermissions": "-rw-rw-rw-",
      "FileType": "FLAC",
      "FileTypeExtension": "flac",
      "MIMEType": "audio/flac",
      "BlockSizeMin": 1152,
      "BlockSizeMax": 1152,
      "FrameSizeMin": 22,
      "FrameSizeMax": 1968,
      "SampleRate": 16000,
      "Channels": 1,
      "BitsPerSample": 16,
      "TotalSamples": 70284,
      "MD5Signature": "d0dddb4a83787e177129050f37177988",
      "Vendor": "reference libFLAC 1.3.0 20130526",
      "Duration": "4.39 s"
    }
  },
  "summary": {
    "file_name": "46fd7184-df87-4a09-a867-1ebfd181f105.flac",
    "file_size": "75249",
    "format": "flac",
    "duration": "4.392750",
    "codec": "flac",
    "sample_rate": "16000",
    "channels": 1,
    "bitrate": "137042",
    "encoder": null,
    "creation_time": "2026:09:17 18:28:25+05:30",
    "modification_time": "2026:09:17 18:28:25+05:30",
    "device_info": null,
    "software": null,
    "comments": null
  },
  "analysis": {
    "findings": [
      {
        "field": "device",
        "has_issue": true,
        "severity": "low",
        "description": "Device information is missing; provenance cannot be verified from metadata alone."
      },
      {
        "field": "encoder",
```

```
        "has_issue": true,
        "severity": "low",
        "description": "Encoder or software information is missing."
      }
    ],
    "risk_score": "low",
    "suspicious_fields": [],
    "missing_fields": [
      "device_info",
      "encoder",
      "software",
      "comments"
    ]
  }
}
```


Raw signal report


```
{
  "spectral_features": {
    "spectral_centroid": {
      "mean": 1220.6410496138178,
      "std": 783.7913829685122,
      "min": 0.0,
      "max": 4683.424306653209
    },
    "spectral_bandwidth": {
      "mean": 1568.035742343826,
      "std": 700.126634283738,
      "min": 0.0,
      "max": 2919.755865543769
    },
    "spectral_rolloff": {
      "mean": 2769.854697452229,
      "std": 1759.809549945914,
      "min": 0.0,
      "max": 6875.0
    },
    "spectral_flatness": {
      "mean": 0.11836665453760778,
      "std": 0.307879006335075,
      "min": 0.00014797002950217575,
      "max": 1.0000007152557373
    },
    "spectral_contrast": {
      "mean": 19.43972139053957,
      "std": 4.39205003465498,
      "min": 3.2286318190080943,
      "max": 28.83425701488692
    }
  },
  "temporal_features": {
    "zero_crossing_rate": {
      "mean": 0.047503234474522295,
      "std": 0.055599724612923104,
      "min": 0.0,
      "max": 0.3115234375
    },
    "rms_energy": {
      "mean": 0.07128062289301602,
      "std": 0.07677072339458452,
      "min": 0.0,
      "max": 0.24804601073265076
    },
    "pitch": {
      "mean": 84.69792099668587,
      "std": 10.157695116219449,
      "min": 68.49972922973525,
      "max": 110.63722351746388
    }
  },
  "anomalies": {
    "anomalies": [],
    "risk_score": "low",
    "summary": "No broad signal-level anomalies identified."
  },
  "temporal_analysis": {
    "consistency_score": 1.0,
    "anomalies": [],
    "summary": "Temporal patterns are broadly consistent.",
    "pitch_analysis": {
      "stability_score": 1.0,
```

```
      "anomalies": [],
      "description": "Pitch contour is broadly continuous."
    },
    "energy_analysis": {
      "energy_mean": 0.07128062844276428,
      "energy_std": 0.07677072286605835,
      "anomalies": []
    }
  },
  "plots": [
    "results\\signal_plots\\waveform.png",
    "results\\signal_plots\\spectrogram.png",
    "results\\signal_plots\\mfcc.png",
    "results\\signal_plots\\pitch_contour.png"
  ],
  "signal_risk": "low",
  "summary": "Signal risk is low. No broad signal-level anomalies identified."
}
```


Methodology

This report compiles the supplied AI, metadata, signal, and fusion JSON outputs. No new model inference is performed during report generation. Evidence sources remain separated to preserve interpretability.